

Norway

SLEIPNER

Volve F

F-11

15/9-F-11 A

Design: 15/9-F-11 A

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	MI @ 54.90m (MI)
Site:	Volve F	MD Reference:	MI @ 54.90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 A	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 A	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site	Volve F, 15/9				
Site Position:		Northing:	6,478,563.53 m	Latitude:	58° 26' 29.8068 N
From:	Map	Easting:	435,050.02 m	Longitude:	1° 53' 14.9285 E
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in	Grid Convergence:	-0.95 °

Well	F-11					
Well Position	+N/-S	4.65 m	Northing:	6,478,568.18 m	Latitude:	58° 26' 29.9566 N
	+E/-W	-0.93 m	Easting:	435,049.09 m	Longitude:	1° 53' 14.8667 E
Position Uncertainty		0.00 m	Wellhead Depth:	91.00 m	Water Depth:	91.00 m

Wellbore	15/9-F-11 A				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	MNETICREFERENCE	31.12.2010	-1.67	71.63	50,439

Design	15/9-F-11 A				
Audit Notes:					
Version:	1.0	Phase:	ACTUAL	Tie On Depth:	2,580.00
Vertical Section:	Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)	
	145.90	4.65	-0.93	23.03	

Survey Program	Date	05.01.2016			
From (m)	To (m)	Survey (Wellbore)	Tool Name	Description	
150.00	2,580.00	F-11 T2 SDI Keeper Gyro (15/9-F-11 T2)	Keeper, cont	Gyro Tool from SDC	
2,590.00	3,144.00	15/9-F-11 A Gyro (15/9-F-11 A)	Keeper, cont	Gyro Tool from SDC	
3,168.10	3,749.10	15/9-F-11 A after Gyro (15/9-F-11 A)	Magn, IFR, mag-corr, dual in-	Magnetic Tools (MWD, EMS)	

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
145.90	0.00	0.00	145.90	4.65	-0.93	0.00	0.000	0.000	0.000	
150.00	0.09	227.76	150.00	4.65	-0.93	0.00	0.659	0.659	0.000	
160.00	0.20	225.57	160.00	4.63	-0.95	-0.03	0.330	0.330	-6.570	
170.00	0.19	227.59	170.00	4.61	-0.97	-0.06	0.036	-0.030	6.060	
180.00	0.23	231.32	180.00	4.58	-1.00	-0.09	0.127	0.120	11.190	
190.00	0.23	233.02	190.00	4.56	-1.03	-0.13	0.020	0.000	5.100	
200.00	0.22	235.12	200.00	4.54	-1.06	-0.16	0.039	-0.030	6.300	
210.00	0.24	234.55	210.00	4.51	-1.10	-0.19	0.060	0.060	-1.710	
220.00	0.34	239.76	220.00	4.49	-1.14	-0.23	0.310	0.300	15.630	

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Site:	Volve F	MD Reference:	MI @ 54.90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 A	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 A	Database:	Production EDM P246N

Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
230.00	0.42	244.41	230.00	4.46	-1.20	-0.29	0.257	0.240	13.950
240.00	0.57	252.51	240.00	4.42	-1.28	-0.35	0.495	0.450	24.300
250.00	0.92	274.57	250.00	4.42	-1.41	-0.40	1.339	1.050	66.180
260.00	1.13	286.19	260.00	4.45	-1.58	-0.44	0.883	0.630	34.860
270.00	1.16	287.83	269.99	4.51	-1.77	-0.46	0.133	0.090	4.920
280.00	1.20	290.33	279.99	4.58	-1.97	-0.48	0.196	0.120	7.500
290.00	1.24	293.03	289.99	4.65	-2.16	-0.48	0.210	0.120	8.100
300.00	1.20	293.20	299.99	4.74	-2.36	-0.48	0.120	-0.120	0.510
310.00	1.19	294.42	309.99	4.82	-2.55	-0.48	0.082	-0.030	3.660
320.00	1.15	292.81	319.98	4.90	-2.74	-0.48	0.155	-0.120	-4.830
330.00	1.12	290.98	329.98	4.98	-2.92	-0.48	0.141	-0.090	-5.490
340.00	1.10	290.84	339.98	5.05	-3.10	-0.49	0.061	-0.060	-0.420
350.00	1.08	288.68	349.98	5.11	-3.28	-0.50	0.137	-0.060	-6.480
360.00	1.07	287.43	359.98	5.17	-3.46	-0.51	0.076	-0.030	-3.750
370.00	1.10	286.47	369.97	5.22	-3.64	-0.53	0.105	0.090	-2.880
380.00	1.19	289.32	379.97	5.29	-3.83	-0.55	0.319	0.270	8.550
390.00	1.33	294.68	389.97	5.37	-4.03	-0.55	0.549	0.420	16.080
400.00	1.73	303.22	399.97	5.50	-4.27	-0.52	1.378	1.200	25.620
410.00	1.86	305.68	409.96	5.68	-4.52	-0.46	0.453	0.390	7.380
420.00	1.89	306.23	419.96	5.87	-4.79	-0.39	0.105	0.090	1.650
430.00	1.96	306.14	429.95	6.07	-5.06	-0.31	0.210	0.210	-0.270
440.00	1.99	306.93	439.94	6.27	-5.34	-0.23	0.122	0.090	2.370
450.00	2.01	307.01	449.94	6.48	-5.62	-0.15	0.061	0.060	0.240
460.00	1.99	307.09	459.93	6.69	-5.89	-0.06	0.061	-0.060	0.240
470.00	2.03	307.66	469.93	6.91	-6.17	0.02	0.134	0.120	1.710
480.00	2.06	308.33	479.92	7.13	-6.45	0.12	0.115	0.090	2.010
490.00	2.05	307.74	489.91	7.35	-6.74	0.21	0.070	-0.030	-1.770
500.00	2.10	310.18	499.91	7.57	-7.02	0.31	0.305	0.150	7.320
510.00	2.11	309.44	509.90	7.81	-7.30	0.41	0.087	0.030	-2.220
520.00	2.12	308.88	519.89	8.04	-7.59	0.52	0.069	0.030	-1.680
530.00	2.09	308.61	529.89	8.27	-7.87	0.62	0.095	-0.090	-0.810
540.00	2.13	310.13	539.88	8.51	-8.16	0.72	0.206	0.120	4.560
550.00	2.13	309.28	549.87	8.74	-8.44	0.83	0.095	0.000	-2.550
560.00	2.12	309.17	559.87	8.98	-8.73	0.93	0.032	-0.030	-0.330
570.00	2.13	309.03	569.86	9.21	-9.02	1.03	0.034	0.030	-0.420
580.00	2.16	309.02	579.85	9.45	-9.31	1.14	0.090	0.090	-0.030
590.00	2.14	309.85	589.85	9.69	-9.60	1.24	0.111	-0.060	2.490
600.00	2.18	309.02	599.84	9.93	-9.89	1.35	0.152	0.120	-2.490
610.00	2.15	308.45	609.83	10.16	-10.18	1.45	0.111	-0.090	-1.710
620.00	2.13	307.49	619.82	10.39	-10.48	1.55	0.123	-0.060	-2.880
630.00	2.01	302.74	629.82	10.60	-10.77	1.62	0.628	-0.360	-14.250
640.00	1.68	284.45	639.81	10.73	-11.06	1.63	2.012	-0.990	-54.870
650.00	1.63	275.56	649.81	10.78	-11.35	1.57	0.784	-0.150	-26.670

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Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 A	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 A	Database:	Production EDM P246N

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
660.00	1.66	272.60	659.80	10.80	-11.63	1.47	0.270	0.090	-8.880	
670.00	1.79	276.80	669.80	10.83	-11.93	1.38	0.544	0.390	12.600	
680.00	2.11	265.05	679.79	10.83	-12.27	1.25	1.532	0.960	-35.250	
690.00	2.45	259.38	689.79	10.77	-12.66	1.04	1.223	1.020	-17.010	
700.00	2.77	256.59	699.78	10.68	-13.11	0.78	1.033	0.960	-8.370	
710.00	3.19	251.91	709.76	10.54	-13.61	0.46	1.455	1.260	-14.040	
720.00	4.01	244.94	719.74	10.30	-14.19	0.01	2.784	2.460	-20.910	
730.00	4.67	241.67	729.71	9.96	-14.87	-0.57	2.114	1.980	-9.810	
740.00	5.38	239.18	739.67	9.53	-15.63	-1.26	2.228	2.130	-7.470	
750.00	6.03	237.66	749.63	9.01	-16.47	-2.07	2.002	1.950	-4.560	
760.00	6.89	236.29	759.56	8.39	-17.42	-3.01	2.621	2.580	-4.110	
770.00	7.68	235.54	769.48	7.68	-18.47	-4.07	2.387	2.370	-2.250	
780.00	8.37	235.75	779.38	6.89	-19.62	-5.25	2.072	2.070	0.630	
790.00	9.33	235.86	789.26	6.03	-20.89	-6.54	2.880	2.880	0.330	
800.00	10.21	236.82	799.12	5.09	-22.30	-7.96	2.685	2.640	2.880	
810.00	10.79	237.99	808.95	4.11	-23.84	-9.46	1.854	1.740	3.510	
820.00	11.33	239.39	818.77	3.11	-25.48	-11.02	1.809	1.620	4.200	
830.00	11.93	242.49	828.56	2.13	-27.24	-12.61	2.598	1.800	9.300	
840.00	12.31	246.06	838.34	1.22	-29.13	-14.19	2.521	1.140	10.710	
850.00	12.52	249.24	848.10	0.41	-31.12	-15.72	2.145	0.630	9.540	
860.00	12.79	252.71	857.86	-0.31	-33.19	-17.18	2.420	0.810	10.410	
870.00	12.84	255.52	867.61	-0.91	-35.32	-18.58	1.876	0.150	8.430	
880.00	12.75	256.87	877.36	-1.44	-37.47	-19.90	0.937	-0.270	4.050	
890.00	12.55	257.29	887.12	-1.93	-39.61	-21.19	0.660	-0.600	1.260	
900.00	12.18	258.31	896.89	-2.38	-41.70	-22.42	1.289	-1.110	3.060	
910.00	11.40	260.19	906.68	-2.77	-43.71	-23.56	2.608	-2.340	5.640	
920.00	10.91	261.87	916.49	-3.07	-45.62	-24.59	1.764	-1.470	5.040	
930.00	10.76	262.94	926.31	-3.32	-47.48	-25.54	0.753	-0.450	3.210	
940.00	10.58	264.05	936.14	-3.53	-49.32	-26.46	0.820	-0.540	3.330	
950.00	10.53	264.50	945.97	-3.71	-51.14	-27.34	0.289	-0.150	1.350	
960.00	10.52	265.36	955.80	-3.87	-52.96	-28.20	0.472	-0.030	2.580	
970.00	10.48	266.01	965.63	-4.01	-54.78	-29.04	0.375	-0.120	1.950	
980.00	10.36	267.69	975.47	-4.11	-56.59	-29.83	0.980	-0.360	5.040	
990.00	10.31	270.33	985.31	-4.14	-58.38	-30.56	1.429	-0.150	7.920	
1,000.00	10.29	272.52	995.14	-4.09	-60.17	-31.22	1.176	-0.060	6.570	
1,010.00	10.34	273.32	1,004.98	-4.00	-61.95	-31.84	0.455	0.150	2.400	
1,020.00	10.39	274.63	1,014.82	-3.88	-63.75	-32.42	0.723	0.150	3.930	
1,030.00	10.48	275.34	1,024.65	-3.72	-65.55	-32.99	0.471	0.270	2.130	
1,040.00	10.47	275.48	1,034.49	-3.55	-67.36	-33.54	0.082	-0.030	0.420	
1,050.00	10.49	275.78	1,044.32	-3.37	-69.17	-34.08	0.174	0.060	0.900	
1,060.00	10.66	277.07	1,054.15	-3.17	-71.00	-34.60	0.874	0.510	3.870	
1,070.00	11.01	278.92	1,063.97	-2.90	-72.86	-35.09	1.480	1.050	5.550	
1,080.00	11.46	281.15	1,073.78	-2.56	-74.78	-35.53	1.876	1.350	6.690	

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Survey Report

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Project:	SLEIPNER	TVD Reference:	MI @ 54.90m (MI)
Site:	Volve F	MD Reference:	MI @ 54.90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 A	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 A	Database:	Production EDM P246N

Survey										
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1,090.00	12.02	283.68	1,083.57	-2.13	-76.76	-35.90	2.282	1.680	7.590	
1,100.00	13.04	287.32	1,093.33	-1.54	-78.85	-36.18	3.869	3.060	10.920	
1,110.00	14.19	289.87	1,103.05	-0.79	-81.08	-36.36	3.891	3.450	7.650	
1,120.00	15.14	291.09	1,112.73	0.10	-83.45	-36.48	2.997	2.850	3.660	
1,130.00	16.29	291.76	1,122.35	1.09	-85.97	-36.55	3.493	3.450	2.010	
1,140.00	18.12	291.94	1,131.91	2.19	-88.72	-36.61	5.492	5.490	0.540	
1,150.00	19.37	291.62	1,141.37	3.38	-91.70	-36.68	3.763	3.750	-0.960	
1,160.00	20.20	291.03	1,150.78	4.61	-94.86	-36.78	2.561	2.490	-1.770	
1,170.00	20.94	290.28	1,160.15	5.85	-98.14	-36.93	2.357	2.220	-2.250	
1,180.00	21.74	288.95	1,169.46	7.07	-101.57	-37.15	2.805	2.400	-3.990	
1,190.00	22.06	288.24	1,178.74	8.26	-105.11	-37.43	1.246	0.960	-2.130	
1,200.00	22.30	287.56	1,188.00	9.42	-108.70	-37.77	1.054	0.720	-2.040	
1,210.00	22.53	286.61	1,197.24	10.54	-112.34	-38.17	1.287	0.690	-2.850	
1,220.00	22.78	285.44	1,206.47	11.60	-116.05	-38.64	1.546	0.750	-3.510	
1,230.00	22.91	284.77	1,215.69	12.61	-119.79	-39.17	0.872	0.390	-2.010	
1,240.00	22.99	284.46	1,224.90	13.60	-123.57	-39.74	0.435	0.240	-0.930	
1,250.00	23.10	284.53	1,234.10	14.58	-127.36	-40.32	0.340	0.330	0.210	
1,260.00	23.02	284.46	1,243.30	15.56	-131.15	-40.90	0.254	-0.240	-0.210	
1,270.00	22.40	284.23	1,252.52	16.51	-134.89	-41.49	1.879	-1.860	-0.690	
1,280.00	22.06	284.22	1,261.78	17.44	-138.56	-42.07	1.020	-1.020	-0.030	
1,290.00	21.49	284.19	1,271.07	18.35	-142.15	-42.63	1.710	-1.710	-0.090	
1,300.00	20.95	283.85	1,280.39	19.23	-145.66	-43.20	1.662	-1.620	-1.020	
1,310.00	20.83	283.60	1,289.73	20.08	-149.13	-43.78	0.448	-0.360	-0.750	
1,320.00	20.85	283.68	1,299.08	20.92	-152.59	-44.36	0.104	0.060	0.240	
1,330.00	21.02	283.73	1,308.42	21.76	-156.06	-44.94	0.513	0.510	0.150	
1,340.00	21.31	283.42	1,317.74	22.61	-159.57	-45.53	0.933	0.870	-0.930	
1,350.00	21.61	283.17	1,327.05	23.45	-163.13	-46.15	0.941	0.900	-0.750	
1,360.00	21.97	282.58	1,336.33	24.28	-166.75	-46.80	1.264	1.080	-1.770	
1,370.00	22.45	281.18	1,345.59	25.06	-170.44	-47.53	2.143	1.440	-4.200	
1,380.00	23.22	278.29	1,354.81	25.71	-174.27	-48.43	4.081	2.310	-8.670	
1,390.00	23.84	276.25	1,363.98	26.21	-178.23	-49.51	3.070	1.860	-6.120	
1,400.00	24.53	274.57	1,373.10	26.60	-182.31	-50.75	2.924	2.070	-5.040	
1,410.00	25.42	271.77	1,382.17	26.83	-186.52	-52.19	4.439	2.670	-8.400	
1,420.00	26.38	269.11	1,391.16	26.86	-190.89	-53.87	4.521	2.880	-7.980	
1,430.00	27.21	266.65	1,400.09	26.70	-195.39	-55.78	4.155	2.490	-7.380	
1,440.00	27.94	264.45	1,408.95	26.34	-200.01	-57.92	3.759	2.190	-6.600	
1,450.00	28.74	261.88	1,417.75	25.77	-204.72	-60.29	4.376	2.400	-7.710	
1,460.00	29.48	260.21	1,426.49	25.01	-209.52	-62.86	3.297	2.220	-5.010	
1,470.00	29.92	259.85	1,435.18	24.15	-214.40	-65.56	1.424	1.320	-1.080	
1,480.00	30.40	259.60	1,443.82	23.26	-219.35	-68.32	1.488	1.440	-0.750	
1,490.00	31.22	259.49	1,452.41	22.33	-224.38	-71.15	2.466	2.460	-0.330	
1,500.00	31.87	259.45	1,460.93	21.37	-229.53	-74.04	1.951	1.950	-0.120	
1,510.00	32.35	259.26	1,469.40	20.39	-234.75	-76.99	1.472	1.440	-0.570	
1,520.00	32.72	259.31	1,477.84	19.39	-240.04	-79.97	1.113	1.110	0.150	

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	MI @ 54.90m (MI)
Site:	Volve F	MD Reference:	MI @ 54.90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 A	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 A	Database:	Production EDM P246N

Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
1,530.00	33.21	259.32	1,486.23	18.38	-245.38	-82.99	1.470	1.470	0.030
1,540.00	33.41	259.40	1,494.58	17.37	-250.78	-86.04	0.614	0.600	0.240
1,550.00	33.61	259.49	1,502.92	16.36	-256.21	-89.09	0.618	0.600	0.270
1,560.00	33.95	259.53	1,511.23	15.34	-261.67	-92.16	1.022	1.020	0.120
1,570.00	34.05	259.37	1,519.52	14.32	-267.17	-95.25	0.403	0.300	-0.480
1,580.00	34.02	258.98	1,527.81	13.27	-272.67	-98.37	0.661	-0.090	-1.170
1,590.00	34.03	258.75	1,536.10	12.19	-278.16	-101.51	0.387	0.030	-0.690
1,600.00	33.97	258.48	1,544.39	11.08	-283.64	-104.67	0.487	-0.180	-0.810
1,610.00	34.00	258.45	1,552.68	9.97	-289.12	-107.85	0.103	0.090	-0.090
1,620.00	34.20	258.47	1,560.96	8.84	-294.61	-111.03	0.601	0.600	0.060
1,630.00	34.38	258.36	1,569.22	7.71	-300.13	-114.23	0.571	0.540	-0.330
1,640.00	34.44	258.17	1,577.47	6.56	-305.66	-117.45	0.369	0.180	-0.570
1,650.00	34.42	257.88	1,585.72	5.39	-311.19	-120.69	0.496	-0.060	-0.870
1,660.00	34.35	257.77	1,593.97	4.20	-316.71	-123.95	0.281	-0.210	-0.330
1,670.00	34.31	257.67	1,602.23	3.00	-322.23	-127.21	0.207	-0.120	-0.300
1,680.00	34.23	257.70	1,610.49	1.80	-327.73	-130.46	0.245	-0.240	0.090
1,690.00	34.23	257.66	1,618.76	0.60	-333.22	-133.72	0.068	0.000	-0.120
1,700.00	34.30	257.68	1,627.03	-0.60	-338.72	-136.98	0.213	0.210	0.060
1,710.00	34.41	257.59	1,635.28	-1.81	-344.23	-140.25	0.363	0.330	-0.270
1,720.00	34.04	257.50	1,643.55	-3.03	-349.73	-143.51	1.120	-1.110	-0.270
1,730.00	33.09	257.75	1,651.88	-4.21	-355.13	-146.71	2.880	-2.850	0.750
1,740.00	32.35	258.27	1,660.30	-5.33	-360.41	-149.82	2.375	-2.220	1.560
1,750.00	31.64	258.71	1,668.78	-6.39	-365.61	-152.82	2.242	-2.130	1.320
1,760.00	30.59	259.30	1,677.34	-7.38	-370.68	-155.71	3.280	-3.150	1.770
1,770.00	29.68	259.79	1,685.99	-8.29	-375.62	-158.48	2.828	-2.730	1.470
1,780.00	29.07	260.04	1,694.70	-9.15	-380.44	-161.16	1.867	-1.830	0.750
1,790.00	28.52	260.12	1,703.46	-9.98	-385.19	-163.78	1.654	-1.650	0.240
1,800.00	27.54	260.26	1,712.29	-10.78	-389.82	-166.33	2.947	-2.940	0.420
1,810.00	26.37	260.65	1,721.21	-11.53	-394.29	-168.77	3.550	-3.510	1.170
1,820.00	25.69	260.79	1,730.19	-12.24	-398.62	-171.12	2.048	-2.040	0.420
1,830.00	25.10	261.16	1,739.22	-12.91	-402.86	-173.39	1.833	-1.770	1.110
1,840.00	24.31	261.30	1,748.31	-13.55	-406.99	-175.59	2.376	-2.370	0.420
1,850.00	23.40	261.24	1,757.45	-14.16	-410.98	-177.72	2.731	-2.730	-0.180
1,860.00	22.79	260.74	1,766.65	-14.78	-414.86	-179.80	1.922	-1.830	-1.500
1,870.00	21.92	260.49	1,775.90	-15.40	-418.61	-181.84	2.626	-2.610	-0.750
1,880.00	20.98	260.58	1,785.21	-16.00	-422.22	-183.81	2.822	-2.820	0.270
1,890.00	20.23	261.02	1,794.57	-16.56	-425.69	-185.68	2.297	-2.250	1.320
1,900.00	19.65	261.39	1,803.97	-17.08	-429.06	-187.48	1.781	-1.740	1.110
1,910.00	19.02	261.79	1,813.41	-17.57	-432.34	-189.21	1.931	-1.890	1.200
1,920.00	18.16	262.11	1,822.88	-18.01	-435.49	-190.85	2.598	-2.580	0.960
1,930.00	17.32	262.45	1,832.41	-18.42	-438.51	-192.41	2.539	-2.520	1.020
1,940.00	16.56	262.81	1,841.97	-18.80	-441.40	-193.89	2.302	-2.280	1.080
1,950.00	15.62	263.53	1,851.58	-19.13	-444.15	-195.27	2.883	-2.820	2.160

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	MI @ 54.90m (MI)
Site:	Volve F	MD Reference:	MI @ 54.90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 A	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 A	Database:	Production EDM P246N

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
1,960.00	14.58	264.78	1,861.24	-19.39	-446.75	-196.53	3.269	-3.120	3.750	
1,970.00	13.75	265.49	1,870.93	-19.60	-449.18	-197.67	2.544	-2.490	2.130	
1,980.00	13.12	265.75	1,880.66	-19.78	-451.50	-198.74	1.899	-1.890	0.780	
1,990.00	12.11	266.00	1,890.42	-19.94	-453.68	-199.74	3.034	-3.030	0.750	
2,000.00	10.90	266.44	1,900.22	-20.07	-455.67	-200.64	3.640	-3.630	1.320	
2,010.00	10.04	267.49	1,910.05	-20.16	-457.48	-201.44	2.643	-2.580	3.150	
2,020.00	9.28	269.00	1,919.91	-20.22	-459.16	-202.14	2.403	-2.280	4.530	
2,030.00	8.48	271.02	1,929.79	-20.22	-460.70	-202.75	2.576	-2.400	6.060	
2,040.00	7.52	273.53	1,939.69	-20.16	-462.09	-203.24	3.064	-2.880	7.530	
2,050.00	6.87	275.05	1,949.61	-20.07	-463.34	-203.64	2.032	-1.950	4.560	
2,060.00	6.28	277.53	1,959.55	-19.95	-464.48	-203.97	1.964	-1.770	7.440	
2,070.00	5.66	282.49	1,969.49	-19.77	-465.50	-204.21	2.418	-1.860	14.880	
2,080.00	5.24	287.86	1,979.45	-19.52	-466.42	-204.34	1.981	-1.260	16.110	
2,090.00	4.77	293.02	1,989.41	-19.22	-467.24	-204.38	1.951	-1.410	15.480	
2,100.00	4.19	297.86	1,999.38	-18.89	-467.94	-204.35	2.076	-1.740	14.520	
2,110.00	3.37	306.50	2,009.36	-18.54	-468.50	-204.25	2.989	-2.460	25.920	
2,120.00	2.89	317.71	2,019.34	-18.18	-468.91	-204.08	2.327	-1.440	33.630	
2,130.00	2.79	328.96	2,029.33	-17.78	-469.20	-203.83	1.696	-0.300	33.750	
2,140.00	2.93	341.18	2,039.32	-17.33	-469.41	-203.50	1.873	0.420	36.660	
2,150.00	3.39	355.71	2,049.30	-16.80	-469.52	-203.04	2.760	1.380	43.590	
2,160.00	3.94	7.56	2,059.28	-16.16	-469.49	-202.45	2.800	1.650	35.550	
2,170.00	4.35	16.27	2,069.26	-15.46	-469.34	-201.74	2.250	1.230	26.130	
2,180.00	4.85	25.12	2,079.22	-14.71	-469.06	-200.94	2.600	1.500	26.550	
2,190.00	5.41	33.13	2,089.18	-13.93	-468.62	-200.06	2.724	1.680	24.030	
2,200.00	6.00	38.53	2,099.13	-13.13	-468.04	-199.09	2.391	1.770	16.200	
2,210.00	6.56	42.36	2,109.07	-12.30	-467.33	-198.05	2.097	1.680	11.490	
2,220.00	7.30	46.42	2,119.00	-11.44	-466.48	-196.92	2.661	2.220	12.180	
2,230.00	8.25	51.54	2,128.91	-10.55	-465.46	-195.71	3.524	2.850	15.360	
2,240.00	9.24	55.20	2,138.79	-9.65	-464.24	-194.40	3.406	2.970	10.980	
2,250.00	10.14	57.43	2,148.65	-8.72	-462.84	-193.00	2.925	2.700	6.690	
2,260.00	11.11	59.85	2,158.48	-7.76	-461.26	-191.50	3.202	2.910	7.260	
2,270.00	12.04	61.78	2,168.27	-6.78	-459.51	-189.91	3.022	2.790	5.790	
2,280.00	12.77	63.25	2,178.04	-5.79	-457.60	-188.26	2.386	2.190	4.410	
2,290.00	13.29	64.43	2,187.78	-4.80	-455.58	-186.55	1.752	1.560	3.540	
2,300.00	13.92	65.47	2,197.50	-3.80	-453.45	-184.80	2.027	1.890	3.120	
2,310.00	14.69	65.98	2,207.19	-2.79	-451.20	-182.98	2.341	2.310	1.530	
2,320.00	15.47	66.14	2,216.85	-1.73	-448.82	-181.08	2.343	2.340	0.480	
2,330.00	16.18	66.32	2,226.47	-0.63	-446.32	-179.09	2.135	2.130	0.540	
2,340.00	16.93	66.49	2,236.06	0.51	-443.71	-177.02	2.255	2.250	0.510	
2,350.00	17.72	66.96	2,245.60	1.69	-440.98	-174.87	2.407	2.370	1.410	
2,360.00	18.61	67.36	2,255.10	2.90	-438.10	-172.63	2.696	2.670	1.200	
2,370.00	19.52	67.72	2,264.55	4.14	-435.08	-170.30	2.753	2.730	1.080	
2,380.00	20.43	68.12	2,273.95	5.43	-431.92	-167.88	2.761	2.730	1.200	

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	MI @ 54.90m (MI)
Site:	Volve F	MD Reference:	MI @ 54.90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 A	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 A	Database:	Production EDM P246N

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
2,390.00	21.25	68.32	2,283.30	6.75	-428.61	-165.38	2.469	2.460	0.600	
2,400.00	21.92	68.35	2,292.60	8.10	-425.20	-162.79	2.010	2.010	0.090	
2,410.00	22.61	68.54	2,301.85	9.50	-421.67	-160.13	2.081	2.070	0.570	
2,420.00	23.43	68.67	2,311.05	10.92	-418.03	-157.39	2.465	2.460	0.390	
2,430.00	24.36	68.72	2,320.20	12.39	-414.26	-154.56	2.791	2.790	0.150	
2,440.00	25.31	68.55	2,329.27	13.92	-410.35	-151.62	2.858	2.850	-0.510	
2,450.00	26.29	68.14	2,338.28	15.53	-406.30	-148.56	2.988	2.940	-1.230	
2,460.00	27.23	67.77	2,347.20	17.22	-402.13	-145.38	2.864	2.820	-1.110	
2,470.00	28.23	67.35	2,356.06	19.00	-397.83	-142.06	3.057	3.000	-1.260	
2,480.00	29.08	67.17	2,364.83	20.85	-393.40	-138.62	2.563	2.550	-0.540	
2,490.00	30.08	67.26	2,373.53	22.76	-388.85	-135.08	3.003	3.000	0.270	
2,500.00	31.02	67.18	2,382.14	24.73	-384.17	-131.44	2.823	2.820	-0.240	
2,510.00	31.64	67.35	2,390.68	26.74	-379.37	-127.71	1.879	1.860	0.510	
2,520.00	32.15	67.45	2,399.17	28.77	-374.49	-123.94	1.538	1.530	0.300	
2,530.00	32.87	67.45	2,407.60	30.83	-369.53	-120.10	2.160	2.160	0.000	
2,540.00	34.06	67.96	2,415.95	32.92	-364.43	-116.18	3.668	3.570	1.530	
2,550.00	35.36	68.34	2,424.17	35.04	-359.14	-112.16	3.954	3.900	1.140	
2,560.00	36.24	68.58	2,432.28	37.19	-353.70	-108.05	2.673	2.640	0.720	
2,570.00	36.93	68.78	2,440.31	39.36	-348.15	-103.89	2.101	2.070	0.600	
2,580.00	37.65	69.24	2,448.26	41.53	-342.49	-99.68	2.316	2.160	1.380	
2,590.00	37.87	66.01	2,456.17	43.86	-336.83	-95.32	5.970	0.660	-9.690	
2,600.00	37.35	64.16	2,464.09	46.43	-331.30	-90.79	3.729	-1.560	-5.550	
2,610.00	36.64	62.42	2,472.08	49.13	-325.92	-86.20	3.795	-2.130	-5.220	
2,620.00	36.09	60.49	2,480.13	51.96	-320.71	-81.55	3.809	-1.650	-5.790	
2,630.00	35.64	58.41	2,488.24	54.94	-315.67	-76.84	3.897	-1.350	-6.240	
2,640.00	35.21	56.92	2,496.38	58.04	-310.77	-72.07	2.894	-1.290	-4.470	
2,650.00	35.61	56.12	2,504.53	61.24	-305.94	-67.24	1.837	1.200	-2.400	
2,660.00	36.53	55.70	2,512.62	64.54	-301.06	-62.30	2.858	2.760	-1.260	
2,670.00	37.66	55.31	2,520.59	67.95	-296.09	-57.21	3.463	3.390	-1.170	
2,680.00	38.88	54.83	2,528.45	71.50	-291.01	-51.96	3.767	3.660	-1.440	
2,690.00	39.94	54.27	2,536.17	75.18	-285.84	-46.54	3.354	3.180	-1.680	
2,700.00	41.34	53.90	2,543.76	79.00	-280.57	-40.97	4.262	4.200	-1.110	
2,710.00	42.46	53.52	2,551.20	82.96	-275.19	-35.22	3.445	3.360	-1.140	
2,720.00	43.55	53.28	2,558.51	87.02	-269.71	-29.34	3.307	3.270	-0.720	
2,730.00	44.36	52.96	2,565.71	91.19	-264.16	-23.33	2.520	2.430	-0.960	
2,740.00	45.45	52.84	2,572.80	95.45	-258.53	-17.21	3.280	3.270	-0.360	
2,750.00	46.51	52.65	2,579.75	99.80	-252.80	-10.97	3.206	3.180	-0.570	
2,760.00	47.27	52.31	2,586.58	104.25	-247.01	-4.61	2.399	2.280	-1.020	
2,770.00	48.09	51.97	2,593.31	108.78	-241.18	1.85	2.573	2.460	-1.020	
2,780.00	49.38	51.74	2,599.91	113.43	-235.27	8.44	3.905	3.870	-0.690	
2,790.00	50.52	51.27	2,606.34	118.19	-229.28	15.17	3.586	3.420	-1.410	
2,800.00	51.28	51.01	2,612.65	123.06	-223.23	22.01	2.359	2.280	-0.780	
2,810.00	52.21	50.85	2,618.84	128.01	-217.14	28.95	2.815	2.790	-0.480	
2,820.00	52.67	50.80	2,624.94	133.02	-210.99	35.96	1.385	1.380	-0.150	

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	MI @ 54.90m (MI)
Site:	Volve F	MD Reference:	MI @ 54.90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 A	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 A	Database:	Production EDM P246N

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
2,830.00	53.08	50.36	2,630.97	138.08	-204.83	43.03	1.619	1.230	-1.320	
2,840.00	53.50	50.18	2,636.95	143.20	-198.67	50.16	1.332	1.260	-0.540	
2,850.00	53.62	49.84	2,642.89	148.37	-192.50	57.33	0.896	0.360	-1.020	
2,860.00	54.26	49.55	2,648.78	153.60	-186.34	64.55	2.045	1.920	-0.870	
2,870.00	55.16	48.96	2,654.55	158.93	-180.15	71.87	3.062	2.700	-1.770	
2,880.00	56.30	48.39	2,660.18	164.39	-173.95	79.32	3.700	3.420	-1.710	
2,890.00	57.85	47.94	2,665.62	169.99	-167.69	86.92	4.786	4.650	-1.350	
2,900.00	59.48	47.58	2,670.82	175.73	-161.37	94.68	4.976	4.890	-1.080	
2,910.00	60.79	47.21	2,675.80	181.60	-154.99	102.58	4.046	3.930	-1.110	
2,920.00	61.84	46.99	2,680.60	187.57	-148.56	110.59	3.203	3.150	-0.660	
2,930.00	62.96	46.59	2,685.23	193.64	-142.10	118.70	3.524	3.360	-1.200	
2,940.00	64.55	46.35	2,689.65	199.82	-135.60	126.93	4.814	4.770	-0.720	
2,950.00	66.06	45.95	2,693.83	206.11	-129.05	135.28	4.659	4.530	-1.200	
2,960.00	66.83	45.86	2,697.83	212.49	-122.47	143.73	2.323	2.310	-0.270	
2,970.00	68.07	45.80	2,701.66	218.92	-115.84	152.24	3.724	3.720	-0.180	
2,980.00	69.69	45.30	2,705.27	225.46	-109.18	160.86	5.057	4.860	-1.500	
2,990.00	70.89	44.74	2,708.64	232.11	-102.52	169.59	3.932	3.600	-1.680	
3,000.00	71.87	44.31	2,711.83	238.87	-95.88	178.41	3.184	2.940	-1.290	
3,010.00	72.93	43.75	2,714.85	245.72	-89.25	187.31	3.560	3.180	-1.680	
3,020.00	74.47	43.11	2,717.66	252.69	-82.66	196.30	4.974	4.620	-1.920	
3,030.00	75.12	42.60	2,720.28	259.76	-76.09	205.38	2.446	1.950	-1.530	
3,040.00	75.04	42.30	2,722.86	266.89	-69.57	214.49	0.902	-0.240	-0.900	
3,050.00	75.15	41.60	2,725.43	274.08	-63.11	223.63	2.056	0.330	-2.100	
3,060.00	75.20	41.08	2,727.99	281.34	-56.73	232.81	1.516	0.150	-1.560	
3,070.00	75.12	40.53	2,730.55	288.66	-50.41	242.02	1.613	-0.240	-1.650	
3,080.00	74.75	40.21	2,733.15	296.01	-44.15	251.23	1.446	-1.110	-0.960	
3,090.00	74.93	40.06	2,735.76	303.39	-37.93	260.46	0.693	0.540	-0.450	
3,100.00	74.99	40.38	2,738.36	310.77	-31.70	269.68	0.944	0.180	0.960	
3,110.00	75.01	40.46	2,740.95	318.12	-25.43	278.90	0.239	0.060	0.240	
3,120.00	75.07	40.51	2,743.53	325.47	-19.16	288.12	0.231	0.180	0.150	
3,130.00	75.14	40.67	2,746.10	332.81	-12.87	297.33	0.509	0.210	0.480	
3,140.00	75.37	40.74	2,748.64	340.14	-6.57	306.55	0.719	0.690	0.210	
3,144.00	75.22	40.75	2,749.66	343.07	-4.04	310.23	1.127	-1.125	0.075	
3,168.10	75.05	41.07	2,755.84	360.67	11.21	332.40	0.439	-0.212	0.398	
3,208.40	72.06	41.84	2,767.25	389.64	36.80	369.07	2.293	-2.226	0.573	
3,248.80	67.68	41.55	2,781.15	417.95	62.02	404.99	3.259	-3.252	-0.215	
3,288.70	63.30	40.68	2,797.70	445.30	85.89	439.49	3.347	-3.293	-0.654	
3,329.40	57.55	40.55	2,817.78	472.15	108.93	473.22	4.239	-4.238	-0.096	
3,369.90	53.32	40.54	2,840.75	497.49	130.60	505.02	3.133	-3.133	-0.007	
3,410.20	49.77	38.28	2,865.81	521.86	150.64	535.28	2.953	-2.643	-1.682	
3,450.30	48.06	37.47	2,892.16	545.71	169.20	564.50	1.358	-1.279	-0.606	
3,490.70	44.78	38.77	2,920.01	568.74	187.26	592.75	2.534	-2.436	0.965	
3,531.10	40.65	38.36	2,949.69	590.16	204.34	619.15	3.074	-3.067	-0.304	

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	MI @ 54.90m (MI)
Site:	Volve F	MD Reference:	MI @ 54.90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 A	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 A	Database:	Production EDM P246N

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
3,571.70	39.98	38.78	2,980.65	610.70	220.72	644.46	0.534	-0.495	0.310	
3,611.10	40.00	39.49	3,010.83	630.34	236.70	668.78	0.348	0.015	0.541	
3,652.30	40.03	39.03	3,042.39	650.85	253.46	694.22	0.216	0.022	-0.335	
3,692.30	39.89	38.19	3,073.05	670.92	269.49	718.96	0.418	-0.105	-0.630	
3,732.80	39.93	38.19	3,104.11	691.35	285.56	744.04	0.030	0.030	0.000	
3,749.10	39.98	38.48	3,116.61	699.56	292.05	754.14	0.357	0.099	0.534	
3,762.00	39.98	38.48	3,126.49	706.05	297.21	762.13	0.000	0.000	0.000	

Casing Points					
Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (in)	Hole Diameter (in)	
202.00	202.00	30"	30.000	36.000	
1,357.70	1,334.20	20"	20.000	26.000	
2,570.68	2,440.85	14"	14.000	17.500	

Design Annotations					
Measured Depth (m)	Vertical Depth (m)	Local Coordinates			
		+N/-S (m)	+E/-W (m)	Comment	
3,762.00	3,126.49	706.05	297.21	Projection to TD	

Checked By: _____ Approved By: _____ Date: _____